## DAEN 427 / ISEN 427 / ISEN 627 - Fall 2026

# Exam 1

*"An Aggie does not lie, cheat or steal, or tolerate those who do."*

By signing below, I state that I will follow the Aggie Code of Honor on this exam.

Name/UIN: _____ Signature: _____

**Instructions:** This is a closed-book exam. You may use pen or pencil, a basic or scientific calculator, and one regular sticky note (standard 3" x 3" *Post-it* or equivalent). Leave the sticky note attached to your exam when you turn it in. No other materials or devices (smartphones, tables, computers, smart-glasses, etc.) are permitted.

Four questions ... DAEN/ISEN 427: 10 points + 1 extra-credit point; ISEN 627: 11 points total.

Good luck! ツ

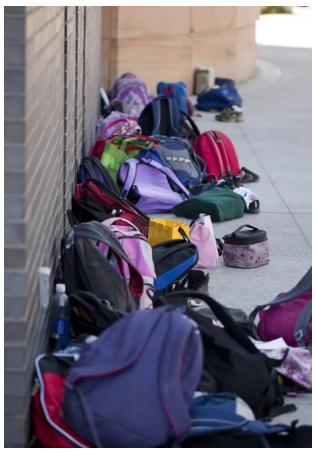
Formulas:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} \qquad P(L_i \mid R) = \frac{P(R \mid L_i)P(L_i)}{\sum_j P(R \mid L_j)P(L_j)}$$

$$\theta \sim \text{Beta}(\alpha, \beta) \;\Rightarrow\; \theta \mid D \sim \text{Beta}(\alpha + x, \beta + n - x) \quad E(\theta \mid D) = \frac{\alpha'}{\alpha' + \beta'} \quad P(\theta > c \mid D) = 1 - F_{\text{Beta}(\alpha', \beta')}(c)$$

$$\mathcal{R}(a_+) = C_{\text{FP}}(1 - q) \qquad \mathcal{R}(a_-) = C_{\text{FN}} q \qquad \tau = \frac{C_{\text{FP}}}{C_{\text{FP}} + C_{\text{FN}}}$$

### Q1. (2 points, 10 min.)



Suppose you are taking a large class with 64 other students. For a class activity, everyone leaves one backpack in a designated storage area before entering the room.

At the end of class, the backpacks are returned one at a time. Your professor reminds everyone: *"A lot of backpacks look alike. Make sure you check carefully before taking one!"*

Good advice: your backpack is a very popular black, medium-sized model used by 15% of students.

From where you are standing, you can see only a backpack's color, size, and general shape. You cannot see its name tag or any other identifying marks until it gets closer.

a. The very first backpack to be returned is black, medium-sized, and looks exactly like your model. What are the chances that it is actually yours? Calculate the posterior probability that the first backpack belongs to you.

b. Fast-forward ... You have now checked the first 63 backpacks and confirmed that none of them is yours. Only two backpacks remain. The 64th backpack appears and, once again, it looks exactly like yours. Now what are the chances that it is yours? Calculate the posterior probability that the 64th backpack belongs to you.

**Answer:**

**Q2. (4 points, 10 min.)**

You have lost your phone somewhere in your home. Based on past experience, the phone is most likely to be in one of four locations: the Bedroom, Living Room, Kitchen, or Office. You use the *"Ring My Phone"* feature and observe whether you can hear the phone ringing.

a. Draw a Directed Acyclic Graph (DAG) for this problem using the following variables:

- historical tendency to lose the phone in different locations;
- location of the phone ($L$); and
- whether you hear the phone ringing ($R$).

Clearly show the direction of the dependencies between the variables.

b. Explain the Bayesian model represented by your DAG. Clearly identify:

- what determines the prior distribution of the location;
- what represents the likelihood of the location given that you hear the phone ringing; and
- what represents the posterior distribution after observing whether the phone can be heard.

c. Suppose the prior probabilities for the phone's location are

$$P(L) = \begin{cases} 0.40 & L = \text{Bedroom}, \\ 0.30 & L = \text{Living Room}, \\ 0.20 & L = \text{Kitchen}, \\ 0.10 & L = \text{Office}. \end{cases}$$

The probability of hearing the phone ring is

$$P(R = 1 \mid L) = \begin{cases} 0.30 & L = \text{Bedroom}, \\ 0.90 & L = \text{Living Room}, \\ 0.60 & L = \text{Kitchen}, \\ 0.20 & L = \text{Office}. \end{cases}$$

You activate *"Ring My Phone"* and hear the phone, so $R = 1$.
Using Bayes' rule,

$$P(L \mid R = 1) = \frac{P(R = 1 \mid L)P(L)}{\sum_{\ell} P(R = 1 \mid L = \ell)P(L = \ell)},$$

calculate $P(L \mid R = 1)$ for each of the four possible locations.

d. Based on the posterior probabilities:

- Where should you search for the phone first?
- Explain how hearing the phone changed your belief about its location compared with the prior probabilities.

**Answer:**

**Q3. (2 points, 10 min.)**

A football defense models the probability $\theta$ that an opposing offense calls a deep pass in a particular down-and-distance situation. The prior is

$$\theta \sim \text{Beta}(3, 12).$$

Game analysis finds 9 deep-pass attempts among 21 comparable plays. The defensive coordinator defines the high-risk state as $H = 1$ when $\theta > 0.40$. The defense may use a conservative two-high safety coverage ($a_+$) or stay with its standard coverage ($a_-$).

a. Find the posterior distribution of $\theta$ and compute $q = P(\theta > 0.40 \mid D)$ (select the correct value from the R outputs below).

b. Suppose the cost of using two-high coverage when the high-risk state is absent is $C_{\text{FP}} = 3$ expected yards, while the cost of staying in standard coverage when the high-risk state is present is $C_{\text{FN}} = 10$ expected yards. Find the threshold, choose the Bayes action, and compare expected losses.

From R, inputs in blue, outputs in red:

```
> 1 - pbeta(0.40, shape1 = 12, shape2 = 24, lower.tail = FALSE) → [1] 0.80483
> 1 - pbeta(0.40, shape1 = 3, shape2 = 15, lower.tail = TRUE) → [1] 0.01232
> 1 - pbeta(0.40, shape1 = 12, shape2 = 24, lower.tail = TRUE) → [1] 0.19518
> 1 - pbeta(0.30, shape1 = 12, shape2 = 24, lower.tail = TRUE) → [1] 0.65156
> 1 - pbeta(0.70, shape1 = 12, shape2 = 24, lower.tail = TRUE) → [1] 2.8536e-06
```

**Answer:**

**Q4. (2 pts + 1 extra pt, 10 min.)**

A student models the probability $\theta$ that their bus will be more than 10 minutes late on a weekday morning. Before tracking this route,

$$\theta \sim \text{Beta}(4, 12).$$

On 16 comparable mornings, the bus was more than 10 minutes late 4 times. Tomorrow, the student can take a rideshare ($a_+$) or take the bus as usual ($a_-$).

a. Find the posterior predictive probability that the bus will be more than 10 minutes late tomorrow.
b. Suppose taking a rideshare when the bus would have been on time costs $C_{\text{FP}} = \$3.00$, while taking the bus when it is late costs $C_{\text{FN}} = \$12.00$ (for example, you will arrive late for your shift and won't get paid for the time, or you will have to make up the hours). Assume all other losses are zero. Find the action threshold and the Bayes action. Verify the decision using expected losses.
c. **(Extra point)** Find the critical false-negative cost above which taking a rideshare becomes optimal, holding $C_{\text{FP}} = \$3.00$ fixed.

**Answer:**